

Problem 4: Best Classifier by ROC Distance

Reading the matrices: rows = predicted, columns = actual, positive class = Positive. TP = predicted Pos & actual Pos, FP = predicted Pos & actual Neg, FN = predicted Neg & actual Pos, TN = predicted Neg & actual Neg.

Step 1: ROC coordinates

Hypothesis	TP	FP	FN	TN	TPR = TP/(TP+FN)	FPR = FP/(FP+TN)
h1	29	1	2	13	29/31 = 0.9355	1/14 = 0.0714
h2	29	3	1	12	29/30 = 0.9667	3/15 = 0.2000
h3	27	3	3	12	27/30 = 0.9000	3/15 = 0.2000

Step 2: Equal costs

Distance from the perfect classifier (FPR = 0, TPR = 1): $d = \sqrt{(FPR)^2 + (1 - TPR)^2}$

Hypothesis	FPR	1 - TPR	Distance
h1	0.0714	0.0645	0.0963
h2	0.2000	0.0333	0.2028
h3	0.2000	0.1000	0.2236

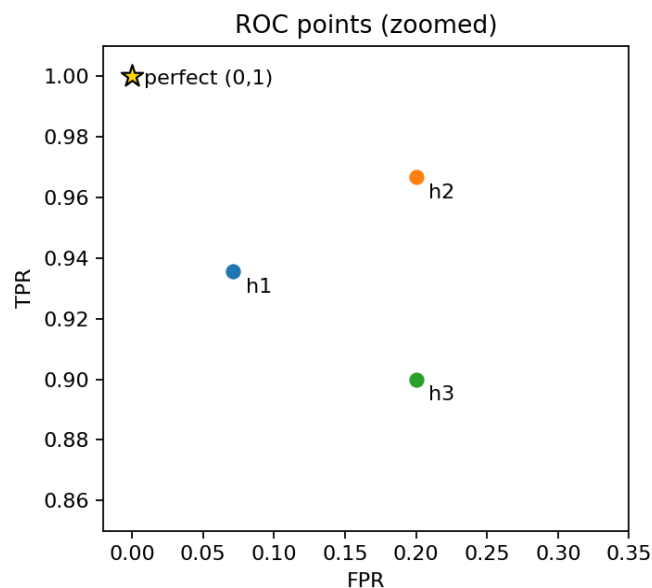
Smallest distance: **h1** (0.0963).

Step 3: False positives cost 4x false negatives

Convention: a false positive is 4 times as costly, so the FPR axis is stretched by a factor of 4: $d_w = \sqrt{(4 \cdot FPR)^2 + (1 - TPR)^2}$

Hypothesis	4 × FPR	1 - TPR	Weighted distance
h1	0.2857	0.0645	0.2929
h2	0.8000	0.0333	0.8007
h3	0.8000	0.1000	0.8062

Smallest weighted distance: **h1** (0.2929).



Answer: (a) Equal costs: h1 is the best classifier. (b) FP cost $4\times$ FN: h1 is again the best. h1 has both the lowest FPR (0.0714) and a TPR close to 1, so it dominates h2 and h3 on FPR and is within 0.03 of h2 on TPR; the heavier FP penalty only widens its lead.